Work Sample:

AI-Assisted Automation of Maintenance Release Notes

Overview

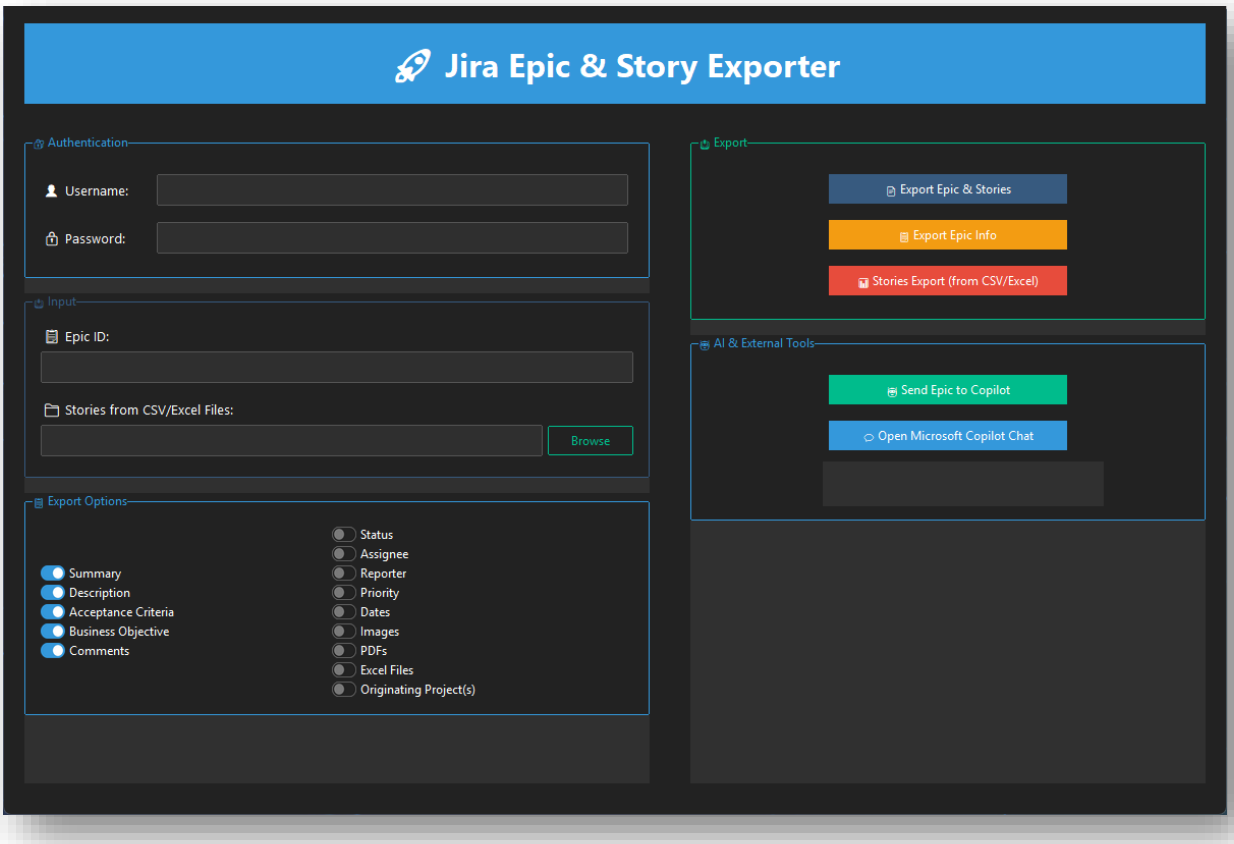
To address the inefficiencies in manual release note creation and to promote productivity gains, I designed and implemented an intelligent documentation workflow that combines Python-based automation with AI-driven content generation.

The existing process required significant manual effort to interpret Jira tickets, restructuring technical content, and ensuring consistency across client deliverables. This typically took **4 to 6 hours per month** and was difficult to scale.

I built a solution that extracts structured Jira data (CSV/XLSX), processes it through an AI system using carefully crafted engineered prompts, and generates client-ready, business-friendly release notes in minutes. The prompts standardize tone, infer impacted modules, and translate technical changes into clear, value-driven summaries.

This approach reduced turnaround time to **under 20 minutes**, while significantly improving consistency and quality. More importantly, it transformed **the generation of Maintenance Release Notes from a manual activity into a repeatable, scalable pipeline.**

The Solution



Tech Stack

Python | **Jira** | **Copilot** | **CSV** | **Prompt Engineering**

Workflow

1. Extract necessary Jira data using a custom Python exporter tool
2. Process data through AI using a structured prompt
3. Generate the desired output as per requirements with all the necessary fields.

Output

Maintenance Release Notes

JIRA ID	Summary	Portal Impacted	Widget Impacted	Notes on JIRA
[illegible]	Need Taxonomy migration feature in EBR	PortalBuilder	Taxonomy Management	Added a new feature to EBR that allows taxonomy changes to be automatically migrated between environments. This eliminates manual effort and improves release efficiency while reducing human error.
[illegible]	Move All Nexgen Infrastructure Provisioning code from SVN to GIT	N/A	N/A	Migrated infrastructure provisioning code to GIT, improving version control, collaboration, and CI/CD integration.
[illegible]	Introduce new lambda event-claims-task for Eventcenter	EventCenter ClaimsCenter	Event Processing	Implemented a lambda function that creates claims tasks automatically based on defined criteria, reducing manual intervention.
[illegible]	Issues on creating change coverage	ESS	Advanced Coverages	Resolved display and processing issues to ensure users can successfully create and manage change coverage.
[illegible]	SecurityCenter Widget Cutover Series: User Access Details	SecurityPlus	User Access Details	Migrated widget from Core Admin to SecurityCenter, improving security management and consistency across the platform.
[illegible]	DSS: Prevent administrators from editing their own security roles	ESS SecurityPlus	Admin Widget	Added a safeguard preventing administrators from modifying their own roles, reducing risk of accidental lockout.
[illegible]	SecurityCenter Widget Cutover Series: Password Change Required	SecurityPlus	Password Change	Enhanced password change functionality for improved user experience and security consistency.

Benefits

- Reduced effort from **4 to 6 hours** to **<20 minutes**, significantly improving turnaround time
- Improved consistency, readability, and tone across all release notes
- Ensured standardized output across multiple clients, eliminating format and quality variations
- Significantly improved client experience and stakeholder confidence through clear, business-friendly communication
- Empowered teams to quickly adapt to changing requirements by simply updating prompts instead of reworking content manually
- Increased documentation agility, allowing faster alignment with client-specific expectations

My Role

- Designed end-to-end solution
- Built Python-based extraction tool
- Engineered AI prompts tailored as per customer requirements
- Defined output structure and validated results.